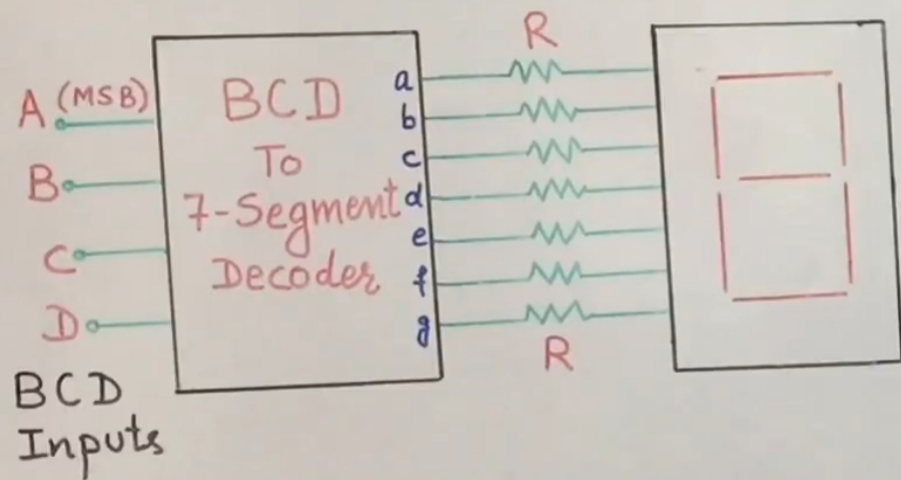
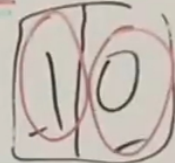
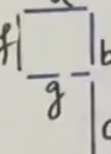
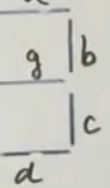
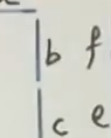
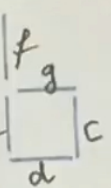
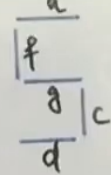
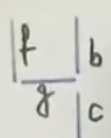
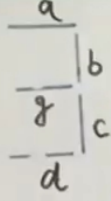
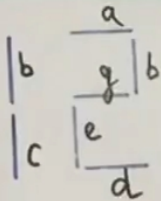
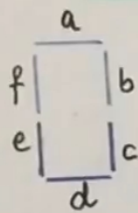


BCD to 7-Segment Display

part-1



	a				a						
Dec Eq.	A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	0							0
1	0	0	0	1	0			0	0	0	0
2	0	0	1	0			0			0	
3	0	0	1	1					0	0	
4	0	1	0	0	0			0	0		
5	0	1	0	1		0			0		
6	0	1	1	0	0	0					
7	0	1	1	1				0	0	0	0
8	1	0	0	0							
9	1	0	0	1				0	0		
10											
15											

Don't care →

BCD to 7-Segment Display

part-2

BCD To 7-SEGMENT DISPLAY DECODER

(a)

AB

00	01	11	10
1	0	X	1
0	1	X	1
1	1	X	X
1	0	X	X

$a = A + CD + \bar{B}\bar{D} + BD$

(b)

AB

00	01	11	10
1	1	X	1
1	0	X	1
1	1	X	X
1	0	X	X

$b = \bar{B} + \bar{C}\bar{D} + CD$

AB

00	01	11	10
1	1	X	1
1	1	X	1
1	1	X	X
0	1	X	X

$c = \bar{C} + B + D$

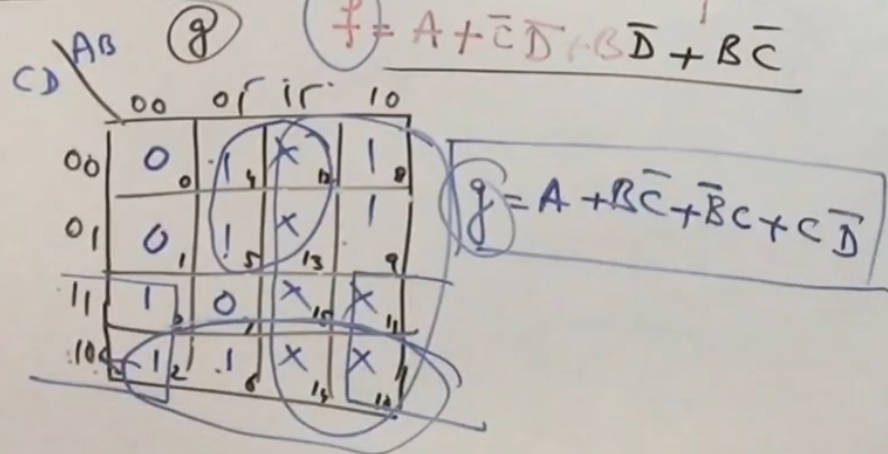
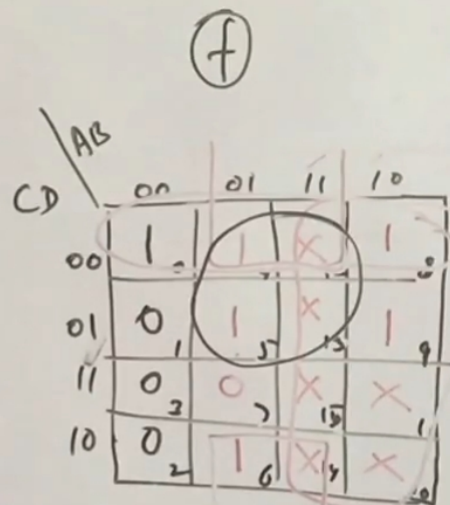
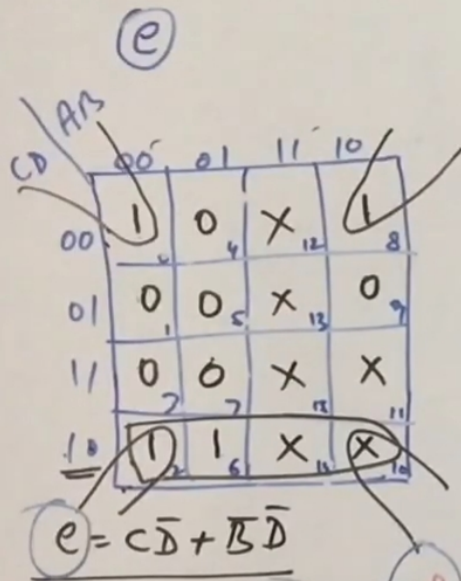
AB

00	01	11	10
1	0	X	1
0	1	X	0
1	0	X	X
1	1	X	X

$d = \bar{C}\bar{D} + \bar{B}C + \bar{B}\bar{D} + BC\bar{D}$

Dec Eq	A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1	0	1	1	0	0	0	0
2	0	0	1	0	1	1	0	1	1	0	1
3	0	0	1	1	1	1	1	1	0	0	1
4	0	1	0	0	0	1	1	0	0	1	1
5	0	1	0	1	1	0	1	1	0	1	1
6	0	1	1	0	0	0	1	1	1	1	1
7	0	1	1	1	1	1	1	0	0	0	0
8	1	0	0	0	1	1	1	1	1	1	1
9	1	0	0	1	1	1	1	0	0	1	1
10	↑ } Don't care										
15											

BCD To 7-SEGMENT DISPLAY DECODER



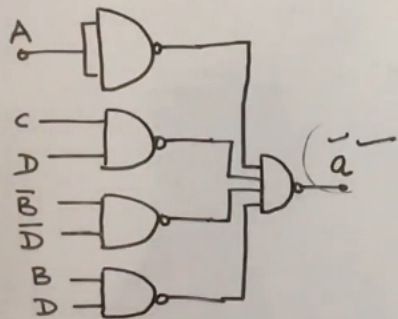
$$\begin{aligned}
 a &= A + \bar{C}\bar{D} + \bar{B}\bar{D} + BD \\
 b &= \bar{B} + \bar{C}\bar{D} + CD \\
 c &= \bar{C} + B + D \\
 d &= \bar{C}\bar{D} + \bar{B}C + \bar{B}\bar{D} + B\bar{C}D \\
 e &= \bar{C}\bar{D} + \bar{B}\bar{D} \\
 f &= A + \bar{C}\bar{D} + \bar{B}\bar{D} + B\bar{C} \\
 g &= A + \bar{B}C + \bar{B}\bar{C} + \bar{C}D
 \end{aligned}$$

Dec Eq	A	B	C	D	a	b	c	d	e
0	0	0	0	0	1	1	1	1	1
1	0	0	0	1	0	1	1	0	0
2	0	0	1	0	1	1	0	1	1
3	0	0	1	1	1	1	1	1	0
4	0	1	0	0	0	1	1	0	1
5	0	1	0	1	1	0	1	1	1
6	0	1	1	0	0	0	1	1	1
7	0	1	1	1	1	1	1	0	0
8	1	0	0	0	1	1	1	1	1
9	1	0	0	1	1	1	1	0	1
10	↑ } Don't care								
11									
12									
13	↑ } Don't care								
14									
15									

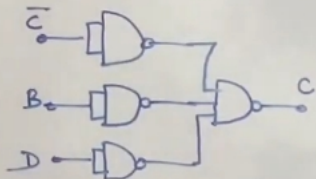
3. Segment display implementation using NAND Gates

BCD To 7-SEGMENT DISPLAY DECODER

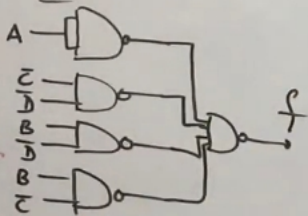
$$(a) = \underline{A} + \underline{C}\underline{D} + \underline{\bar{B}}\underline{\bar{D}} + B\underline{D}$$



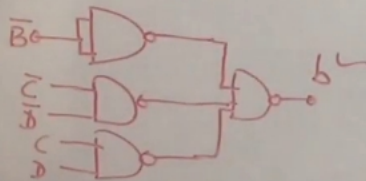
$$C = \underline{\bar{C}} + \underline{B} + \underline{D}$$



$$f = A + \underline{\bar{C}}\underline{\bar{D}} + B\underline{\bar{D}} + B\underline{\bar{C}}$$



$$(b) = \underline{\bar{B}} + \underline{\bar{C}}\underline{\bar{D}} + \underline{C}\underline{D}$$



$$\begin{aligned} a &= A + \underline{C}\underline{D} + \underline{\bar{B}}\underline{\bar{D}} + B\underline{D} & e &= \underline{C}\underline{\bar{D}} + \underline{\bar{B}}\underline{\bar{D}} \\ b &= \underline{\bar{B}} + \underline{\bar{C}}\underline{\bar{D}} + \underline{C}\underline{D} & f &= A + \underline{\bar{C}}\underline{\bar{D}} + B\underline{\bar{D}} + B\underline{\bar{C}} \\ \underline{c} &= \underline{\bar{C}} + \underline{B} + \underline{D} & g &= A + B\underline{\bar{C}} + \underline{\bar{B}}\underline{C} + \underline{C}\underline{D} \\ d &= \underline{C}\underline{\bar{D}} + \underline{\bar{B}}\underline{C} + \underline{\bar{B}}\underline{\bar{D}} + B\underline{\bar{C}}\underline{D} \end{aligned}$$

